Measuring the Specific Heat Capacity of Water

Lab report by Lorcan Nicholls, 28th September 2019

Abstract

The *specific heat capacity (SHC)* is a bulk quantity of a material defining the energy required to raise 1 kg of the substance by 1 °C. In this report, a method for determining the SHC of water is presented with the result and uncertainty analysed. The value obtained was 5100 ± 1300 J kg^{-1} K^{-1}. According to an online source[1], the true value is 4180 J kg^{-1} K^{-1}, which lies within our range.

Introduction

Thermal energy is the kinetic energy of the vibrating particles in a material. Thermal energy transfers are governed by the laws of thermodynamics: the first law of which, states that the sum of the thermal energy changes in a closed system is zero. This is a direct result of conservation of energy, applied to thermal energy.

Theory

When E joules of thermal energy is transferred to a material of mass m kg and specific heat capacity c J kg^{-1} K^{-1}, the increase in its temperature ΔT is related by

$E = mc\Delta T$ (Equation 1).

When the thermal energy is provided by a resistor dissipating electric current I across a potential difference V, the energy transferred in time t is given by

$E = IVt$ (Equation 2)

Equating (1) and (2),

$IVt = mc\Delta T$

If ΔT is taken as the dependent variable while t is varied, and all other quantities are controlled (i.e. kept constant), then the equation can be rewritten,

$mc = IVt/\Delta T = \Delta E/\Delta T$ (Equation 3)

(3) shows c as being proportional to the gradient of a graph of energy against temperature increase, which allows the experimental determination of c by continuous heating and measuring at specified time intervals.

Heat loss to the surroundings is minimised by using a thermodynamic beaker to insulate the water.

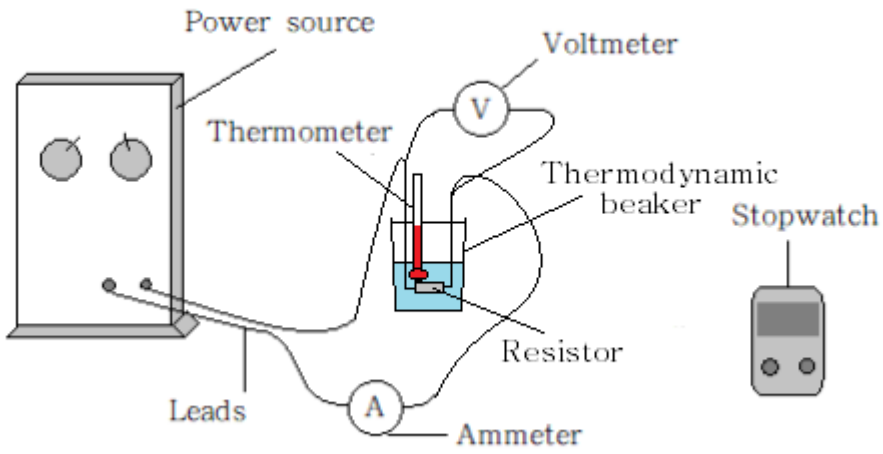
Setup and Method

The apparatus we used was:

- Water (100 grams)
- Voltmeter and ammeter
- Thermodynamic beaker
- Resistor, connected to lid
- Electrical wires
- Power supply
- Thermometer
- Timer

The setup used is shown in *Figure 1*.

Figure 1



Once set up, the procedure we followed was:

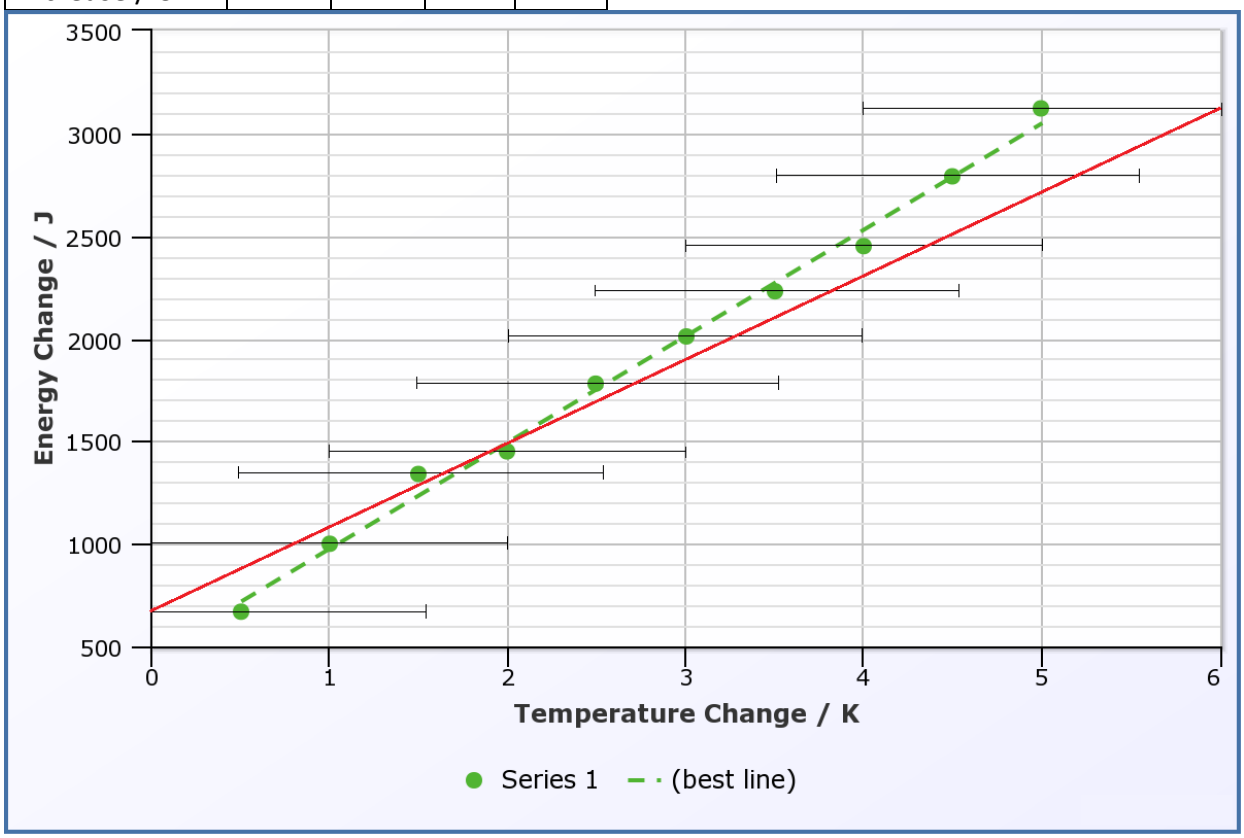
1. Measure the initial temperature of the water
2. Turn the power supply on, with the voltage set to the maximum
3. Start the stopwatch
4. Record the mean value of the p.d. and current with the voltmeter and ammeter
5. When the stopwatch reaches 30 seconds, measure the temperature on the thermometer.
6. Repeat taking measurements from the thermometer every 30 seconds until 15 minutes have passed.
7. Calculate the power of the supply.
8. Plot a graph of results, energy against temperature increase, and draw best fit line
9. Calculate the gradient and hence obtain a value for the SHC of water.

Results

The temperature difference between the initial and at measurement was calculated at regular 30 second intervals for 14 minutes. The results for which a temperature increase was observed are shown in *Table 1*.

Table 1

Time / s	0	60	120	180	270	360	390	480	540
Energy transfer / J	0	223	446	669	1000	1340	1450	1780	2010
Temperature increase / C	0	0	0	0.5	1.0	1.5	2.0	2.5	3.0
Time / s	600	660	750	840					
Energy transfer / J	2230	2450	2790	3121					
Temperature increase / C	3.5	4.0	4.5	5.0					



The best fit line extends from energy changes of 719 J to 3050 J over a temperature of 4.5. This gives a mean gradient of 519 J K^{-1}, leading to a mean SHC value of 5070 J kg K^{-1}. The worst fit line has a gradient of 389, leading to a worst-case SHC of 3800 J kg K^{-1} and therefore a final range of c = (5100 ± 1300) J kg K^{-1}.

The largest source of uncertainty was by far due to the low resolution of the thermometer and the small temperature range which was measured.

The uncertainty in the energy change is negligible compared to the temperature as the voltmeter and ammeter used had total percentage uncertainties of less than 2% throughout. The first three results were omitted from the graph as no temperature change was observed in this period.

Conclusions

We obtained a range for *c* of (5100 ± 1300) J kg K^{-1}, a bound within which the reported true value of 4180 J kg K^{-1} lies within. Therefore, we claim the result is accurate and valid. The precision of this experiment could be improved by automating the process of recording the temperature, using a digital thermometer and having a computer record the data instead of a human experimenter. We could also have used a longer time scale to heat the water, which would have decreased the percentage uncertainty due to the thermometer at the expense of time.

References

[1] The Engineering Toolbox: *Specific heat of water* (Accessed 28th September 2019)

https://www.engineeringtoolbox.com/specific-heat-capacity-water-d_660.html